

Maya Winland-Gaetz

203-1870 McKenzie Avenue, Victoria BC V8N 4X3
mwinlandgaetz@gmail.com, (289) 689-0427

Education	Bachelor of Engineering (BEng) , In Progress, GPA 8.61/9 Expected Sept 2028 Electrical Engineering University of Victoria , Victoria, BC, Canada
	PhD Student , (Incomplete), GPA 11.0/12 September 2013 - May 2019 McMaster University , Hamilton, ON, Canada
	Bachelor of Science (BSc) , GPA 3.57/4 September 2008 - April 2013 Specialist Environmental Toxicology, Maj. Ecology & Evolutionary Biology University of Toronto , Toronto, ON, Canada
Competencies	Technical • Drafted, simulated, and am currently in fabrication for several electronics projects using KiCAD, LTSpice, and Altium Designer. • Developed several applications in C++ for personal use. • 90WPM typing speed with a keen eye for typographical and grammatical errors, using both Microsoft Office and LaTeX. • Prepared and analyzed sensitive geochemical samples in a laboratory environment, implementing state-of-the-art instruments while observing rigorous safety protocols.
	Core • Maintained efficiency while under pressure during the annual Starbucks Holiday Launch, providing a high standard of service for eager early-morning customers. • Communicated clearly and fluently in presentations and public speaking, earning an award at an academic conference for a 15 minute talk on geochemical sampler design.
Projects	First Year Engineering Design Project January 2024 - April 2024 • Conceived and constructed a physical prototype and associated webapp for a smart shower heater as part of a class project. The product could sense the proximity of a user and automatically adjust the water temperature using a hypothetical inline water heater to a desired temperature, configurable through a WiFi-enabled app. • Went beyond the expectations of the course by building transistor amplifiers to facilitate proximity detection with inexpensive IR sensors, and improved the TA-provided netcode to ensure a stable connection. • Worked in a group of 3, wherein the circuit design and netcode improvements were my chief responsibilities. <i>Demonstration video available upon request.</i>
	University of Victoria Satellite Design Club January 2024 - Present • Researched satellite radiocommunication design principles and protocols to guide the development of the telemetry systems for future club projects. • Currently responsible for the development of the Telemetry, Tracking & Command (TT&C) boards for several ongoing projects, including frequency selection,

component research, schematic capture, and acquisition of the required certifications to legally operate in amateur radio frequencies.

- Designed, compiled BOMs, and ordered PCBs for evaluation boards using Altium Designer, through commercially-available fabrication workshops.
- Hand-soldered 7mm QFN and 0603/0402-sized components for the board using a hot air rework station and manual soldering iron to assemble the first iteration of transceiver evaluation boards for testing.

Personal

April 2020-Present

- Currently building a trigger signal sequencer as part of a larger project to construct a music synthesizer, utilizing RP2040 hardware timers and digital logic ICs to output a 4-channel beat structure with a custom LED matrix.
- Programmed and assembled a climate and light level sensor for personal use, incorporating an AtMega328P microcontroller and a scrolling 16x8 LED display.
- Programmed an app in C++ to view a directory of life drawing references at regular intervals, to aid in life drawing study during the pandemic.

Professional Experience

Barista

April 2019 - December 2021

Starbucks, Hamilton, ON, Canada/Saanich, BC, Canada

- Demonstrated clear communication and professional affect in connecting with customers amidst an extremely busy work environment.
- Prepared food and beverages to precise standards of quality through fine attention to detail, prompting glowing remarks on cappuccino quality in particular on several occasions from grateful patrons.
- Achieved role of Certified Barista Trainer through demonstrating a keen interest in teaching others, and modelling workplace standards; responsibilities included structured and hands-on training.

Teaching Assistant

September 2013 - April 2018

McMaster University, Hamilton, ON, Canada

- Planned and conducted lab sessions and tutorials, using previous experience and WHMIS training in chemistry lab safety and analytical protocols, to minimize accidents, control spills and ensure a safe learning environment.
- Graded and annotated student papers and projects on 7-day deadlines, using rigorous organization and uniform standards, to ensure course leaders could maintain a strict schedule.

Awards and Certifications

- Amateur Radio Operator Certification December 2024
Advanced, Basic with Honours (VA7MWX)
- Best Student Oral Presentation, worth \$750 (*Sudbury Mining and Environment Int'l Conference*) June 2015
- WHMIS Yearly Recertification 2008-2022
 - Other safety certifications as needed (Hazardous materials disposal and compressed gases)

Other Interests

- Bicycling, with the goal of getting from the University of Victoria to Leechtown and back over a two day trip.
- Classical piano, playing Beethoven's later Sonatas in particular.

References Available Upon Request